

Monthly Progress Update (April 2026) & May Goals

Work Completed in April 2026:

- Implemented and developed a strong understanding of Keccak algorithm
- Developed a UVM testbench for Keccak verification
- Conducted literature review focusing on recent (2026) research papers related to post-quantum cryptography and hardware implementations

Goals for 2nd Week of May:

- Complete verification of Keccak implementation
- Begin implementation of Number Theoretic Transform (NTT)
- Explore recent research and potential optimizations to improve Keccak and/or Dilithium implementations

Overall Goals Before Midyear Evaluation:

- Complete verification of Keccak
- Complete NTT implementation
- Synthesize Keccak on Intel Quartus
- Strengthen literature review by identifying new optimization approaches from recent research
- Begin structured thesis writing